

Calcutta University
MSC 2nd Year
Advanced Database Management System (CSMC 201)
MID SEM 2024
Full Mark=20

1 Answer any four Question:

[4x2=8]

- a) What is insertion anomaly? Explain with an example.
- b) Define transitive functional dependency with the help of suitable example.
- c) Differentiate between B-Tree and B+ Tree.
- d) Define query optimization.
- e) Find out candidate key for relation $R = (A, B, C, D, E)$ and given functional dependencies are: $FD = \{B \rightarrow CD, D \rightarrow E, B \rightarrow A, E \rightarrow C, AD \rightarrow E\}$.

2 Answer any Three question:

[3x4=12]

- a) Write properties of B-Tree. Insert following elements in the B-Tree of order 4:
65, 66, 70, 71, 74, 80, 91, 81, 99, 82, 75, 77, 89, 56
- b) Explain extendible hashing with the following elements:
16, 6, 4, 22, 24, 10, 31, 7, 9, 20, 26, 28
- c) Explain Cost-Based query optimization for SELECT operation.
- d) Write rule of 3NF. Consider the relation $R = (A, B, C, D, E, F, G, H, I, J)$ and the Functional dependencies are following:
 $FD = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
Decompose R into 3NF.